Carle Illinois
COLLEGE OF MEDICINE

Anomalous Artificial Automatons

Carle
UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

ETHAN, MARK, PRANEETHA, JACKSON, JERRY

Carle Illinois COLLEGE OF MEDICINE

AGENDA

Meet the Team

User Story

Overall Structure

Demonstration Video

Closing Thoughts

Carle Illinois
COLLEGE OF
MEDICINE

THE ARDS AI TEAM

Ethan Man

Rising Senior in
Computer Science
UIUC

Mark Lu

Rising Senior in
Physician-Engineer Program
NYCU

Praneetha Bhogi

Rising Senior in
Computer Science + Philosophy
UIUC

Carle Illinois
COLLEGE OF
MEDICINE

THE LUMBAR PUNCTURE TEAM

Jackson Song

Rising Senior in
Information Sciences
+ Data Science
UIUC

Jerry Ni

Rising Junior in
Computer Science +
Statistic
UIUC

AI in Medical Education

Carle Illinois COLLEGE OF MEDICINE

Practice Anywhere

Learn Anytime

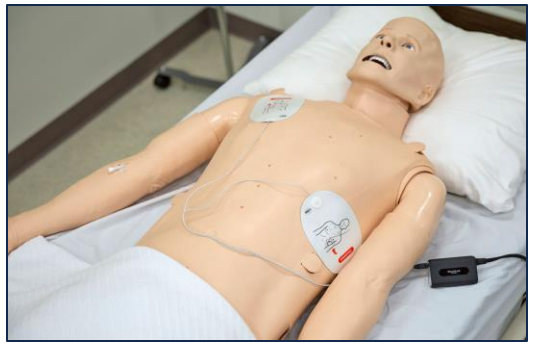
ARDS Project

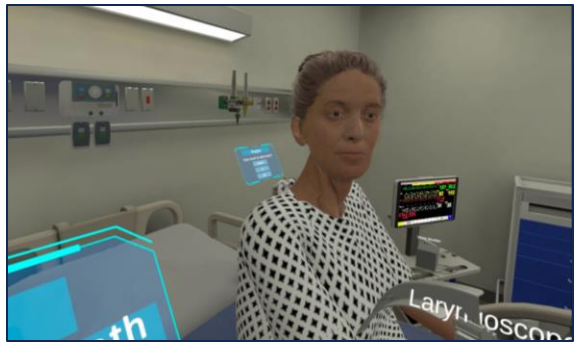
Carle Illinois COLLEGE OF MEDICINE

THE BIGGER PICTURE

BEFORE

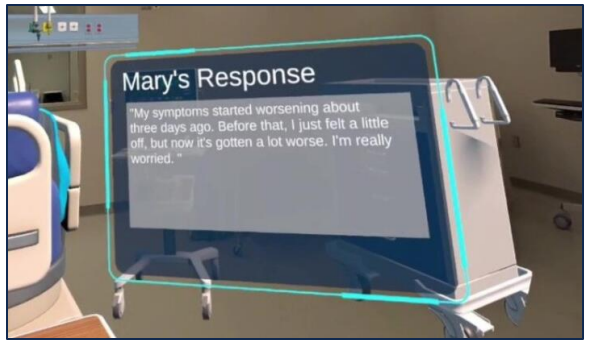
- Manikin simulation: Patient-user dialogue driven by facilitator
- CHF VR simulation: Patient outputs are limited





AFTER

- Unique patient-user dialogue driven by user
- Reduced wait time for answer generation
- Simplistic start-up



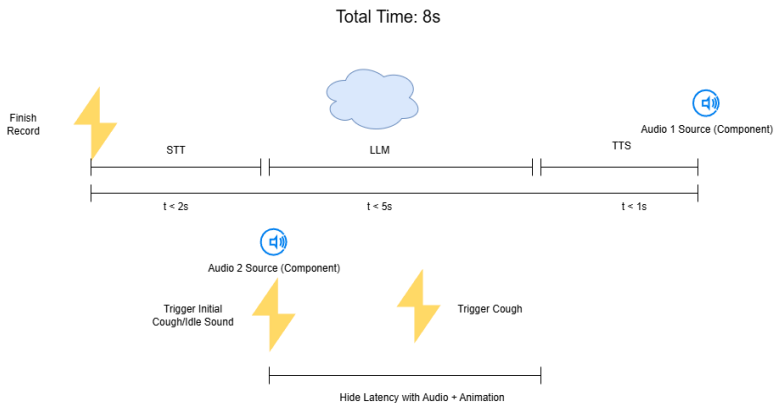
REQUIREMENTS & USER STORIES

Carle Illinois COLLEGE OF MEDICINE

Conversational AI

Real & On-topic

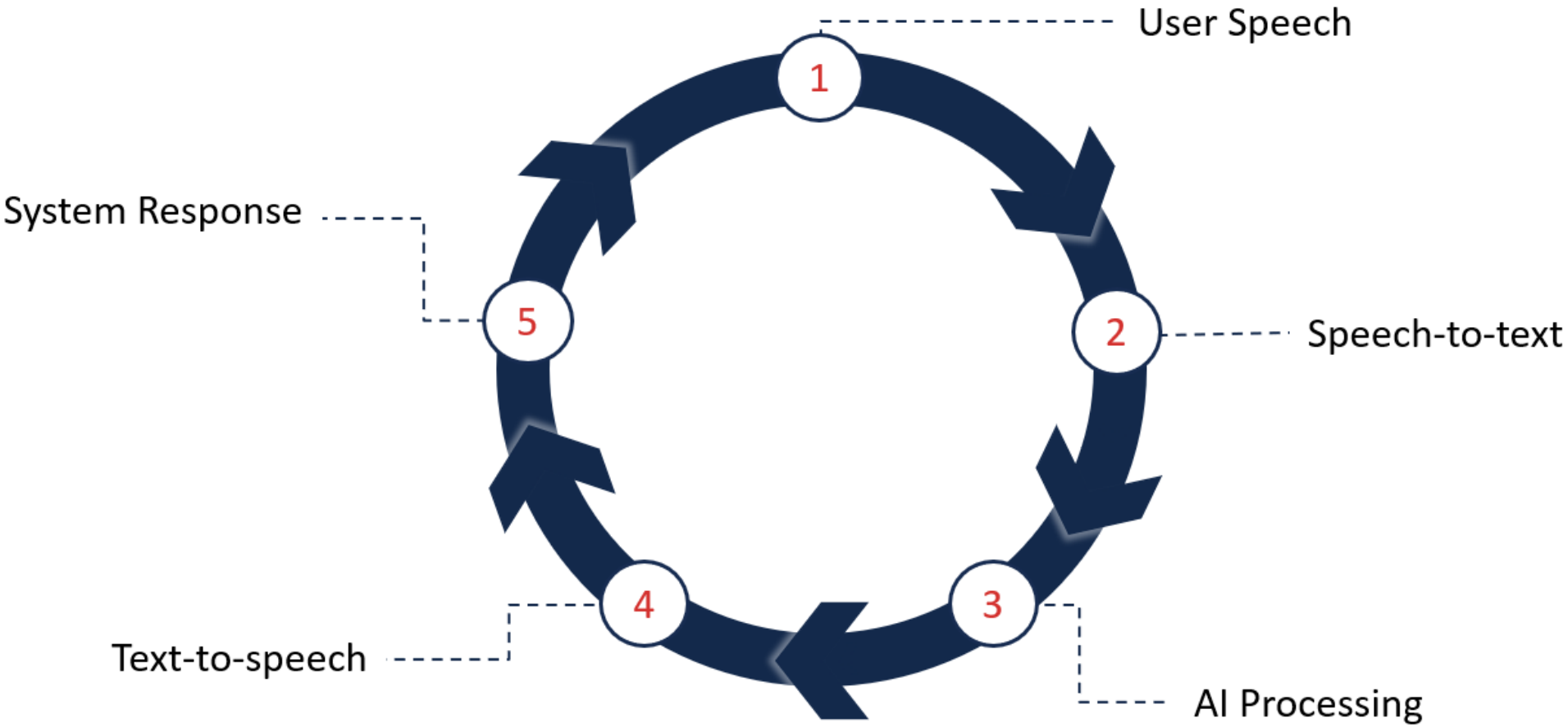
Low Latency



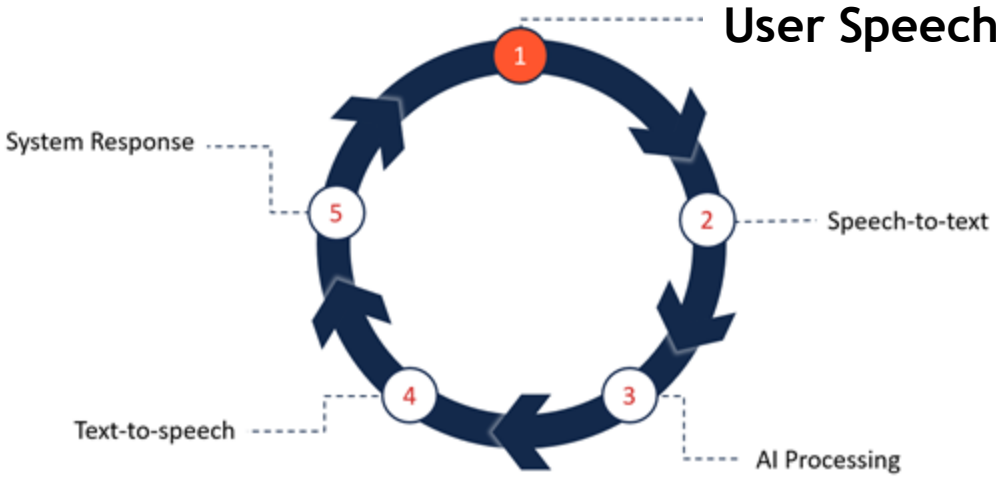
STRUCTURE

OVERALL STRUCTURE

Carle Illinois COLLEGE OF MEDICINE

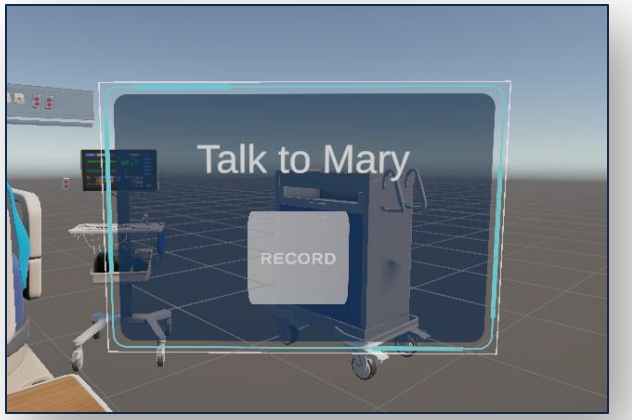


USER SPEECH

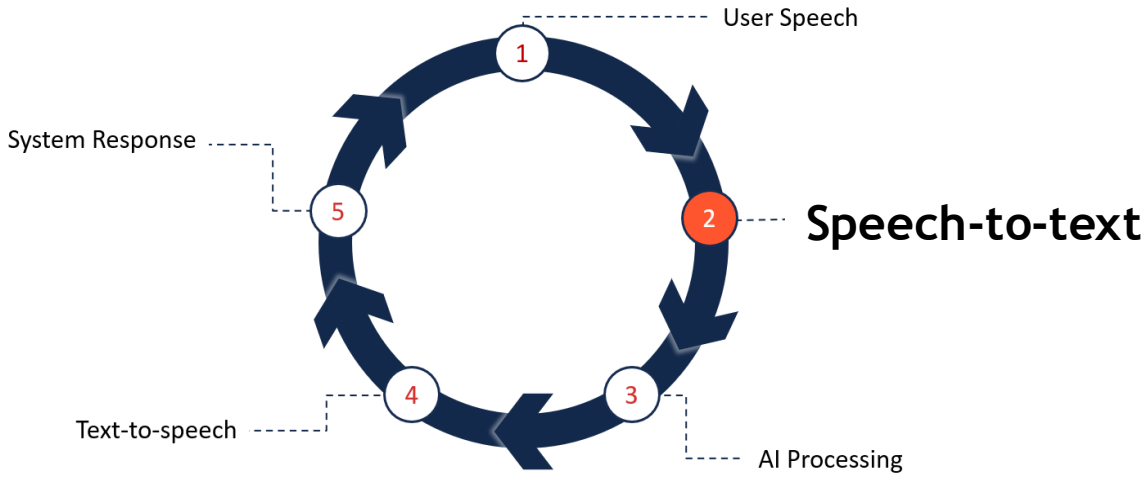


Features

- Clear and concise user interface
- Accepts any question



SPEECH-TO-TEXT



Function

- Translate human audio to text

Features

- High accuracy and speed
- Calls Whisper cloud services

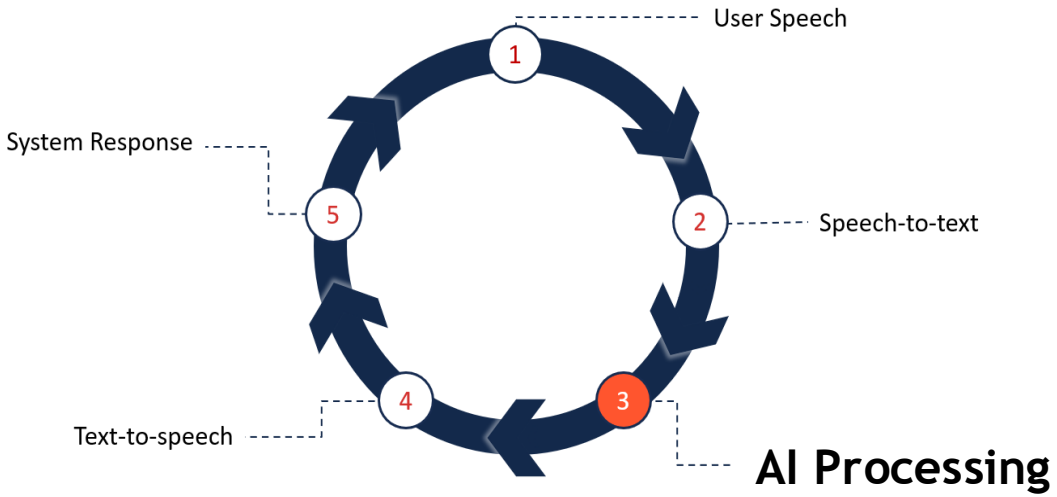
Tools

OpenAI/Whisper

Considerations

IIElevenLabs

AI PROCESSING



Function

- Comprehend and generate messages

Features

- Refer to question-answer history for increased accuracy
- Fall-back answers for blank or off-topic prompts

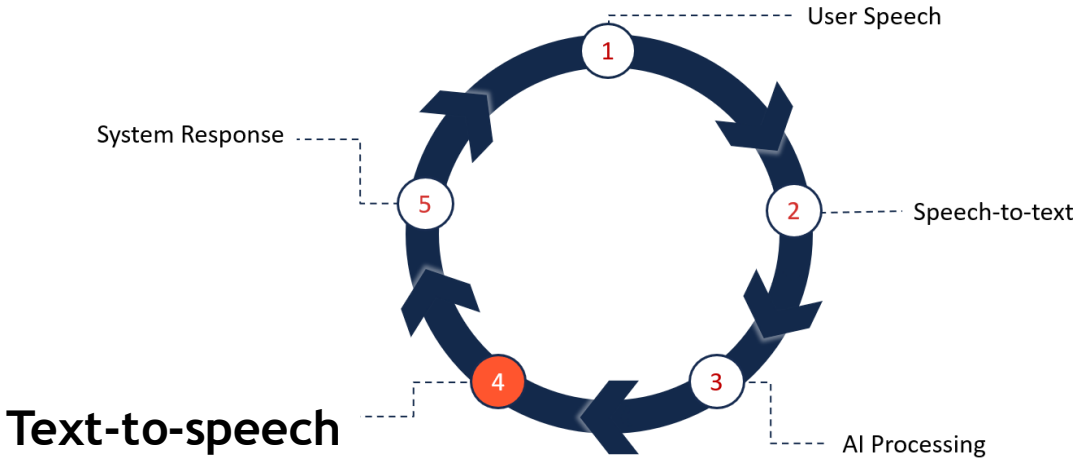
Tools

GPT-4o mini

Amazon DynamoDB

AWS Lambda

TEXT-TO-SPEECH



Function

- Transform AI-generated response into audio

Previous Implementation

- Pydub for customizable splicing of coughing
- Con: High latency
  - Using built-in Unity features instead

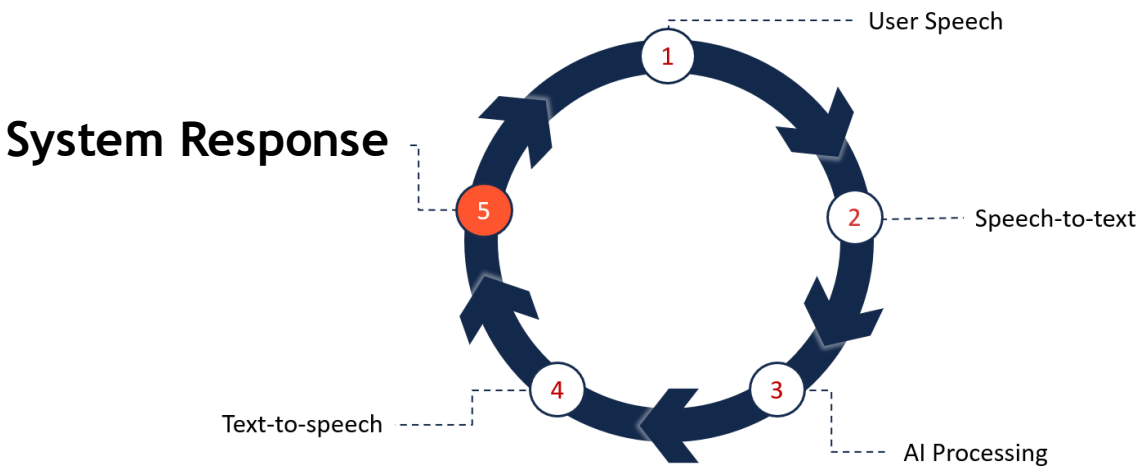
Tools

Lingotion®

Considerations

sesame

SYSTEM RESPONSE

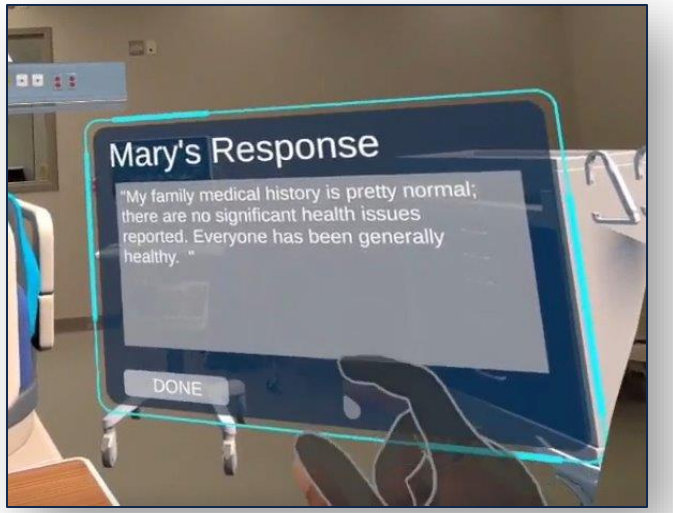


Function

- Play audio response in Mary's voice

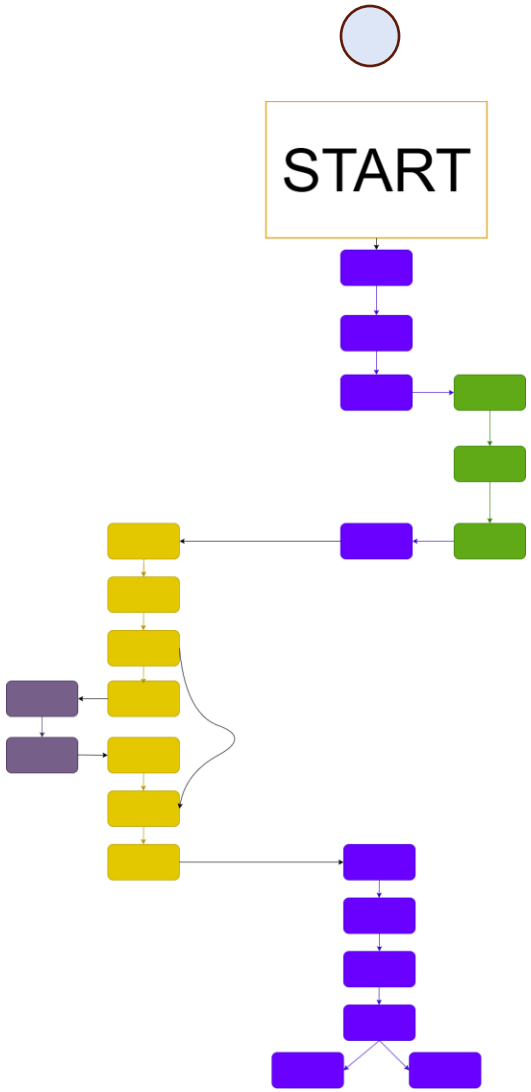
Features

- Various emotions and physical states
- Response appears as visible text as well

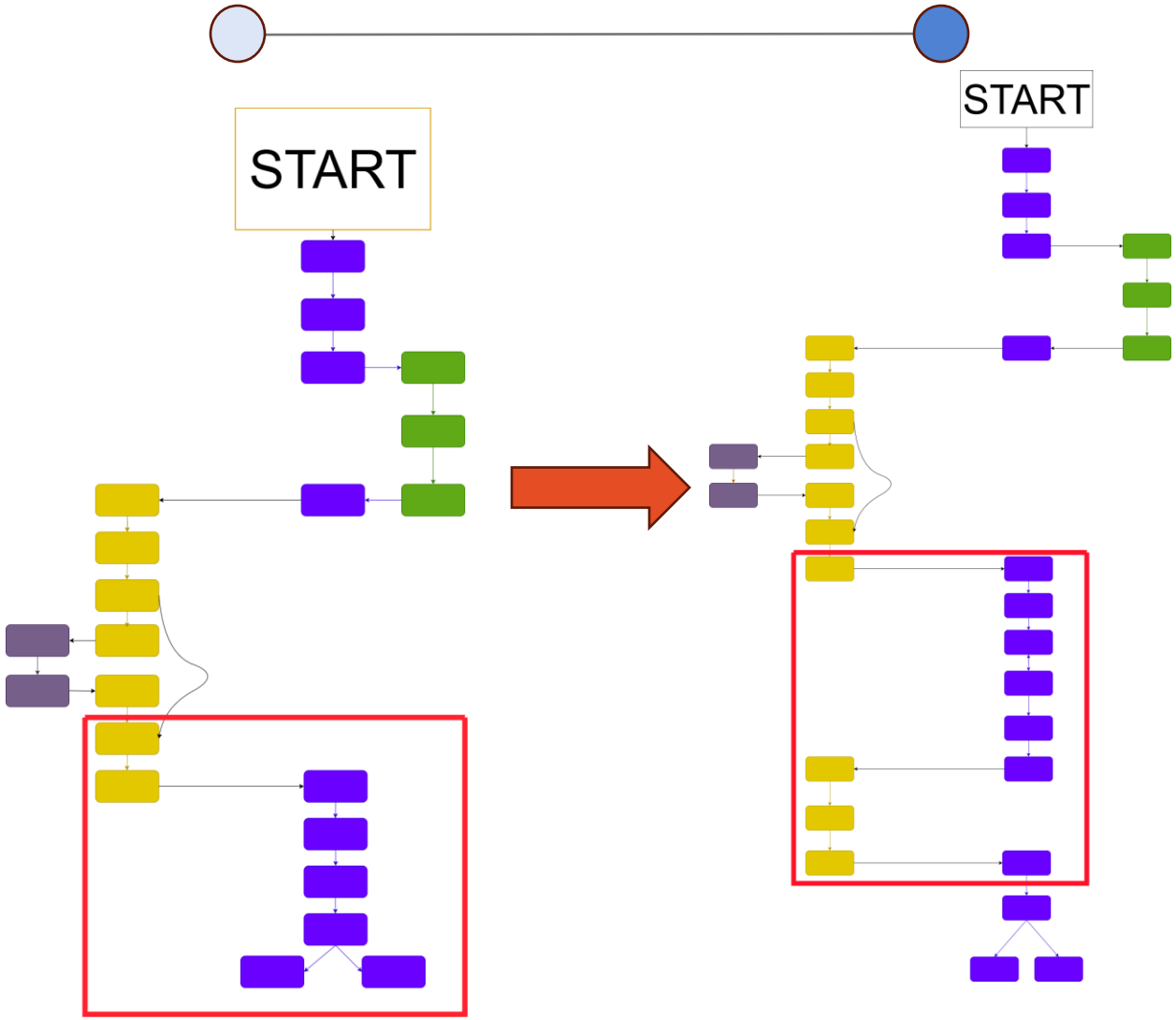


ITERATIVE PROCESS

Carle Illinois COLLEGE OF MEDICINE

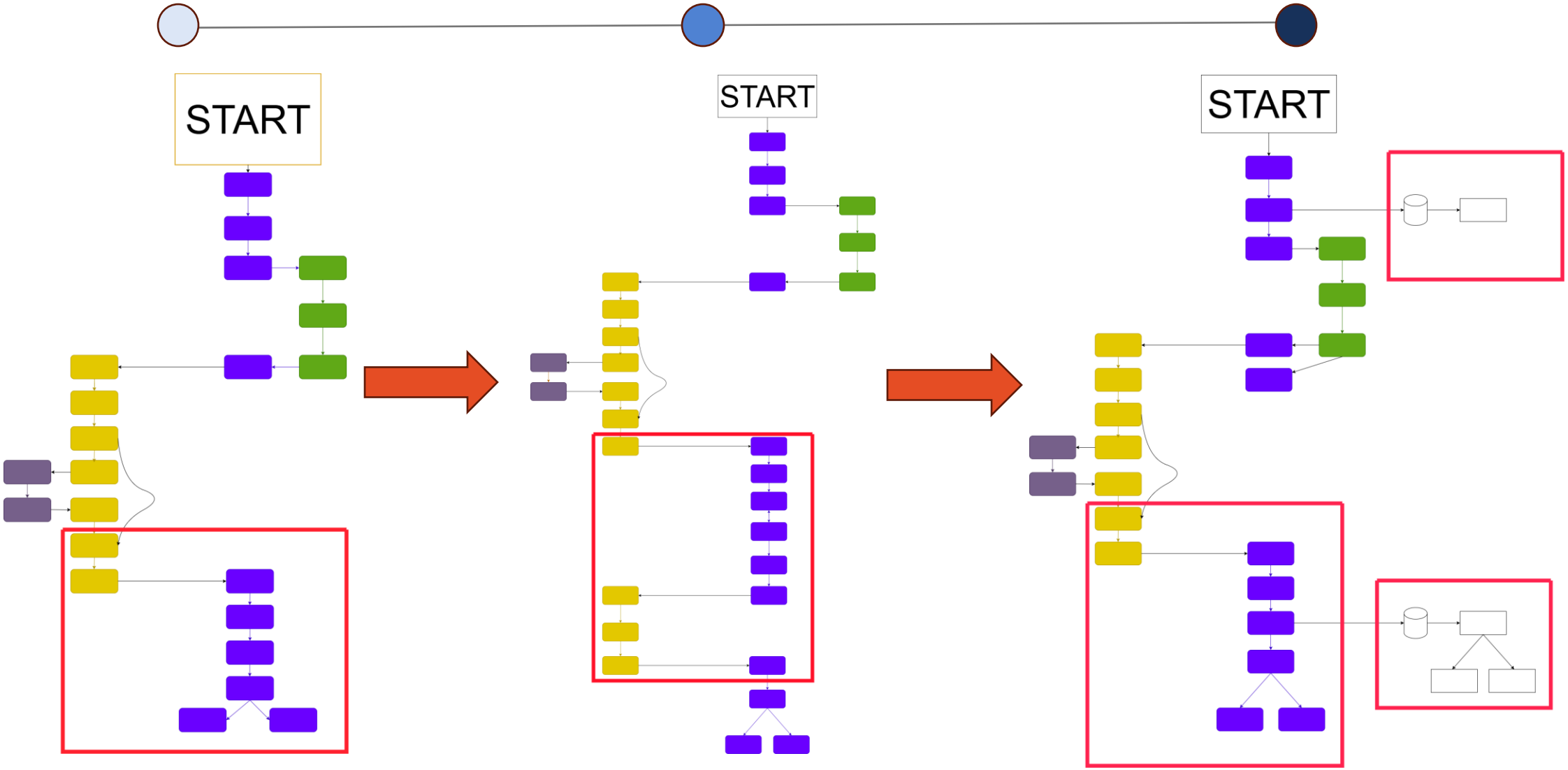


ITERATIVE PROCESS



ITERATIVE PROCESS

Carle Illinois COLLEGE OF MEDICINE



CONCLUSION

Closing Thoughts

Future Steps

- Allow responses to more accurately show progression of symptoms
- Add more splicing to show natural health deterioration

Takeaways

- Connecting multiple systems is complicated
- AI & XR are becoming more useful in medical education

Lumbar Puncture AI Facilitator

Why Lumbar Puncture?

1. It requires procedural skills
2. It relies on task trainers
3. It demands muscle memory

Why we need an AI facilitator in XR Procedural Training?

Carle Illinois COLLEGE OF MEDICINE

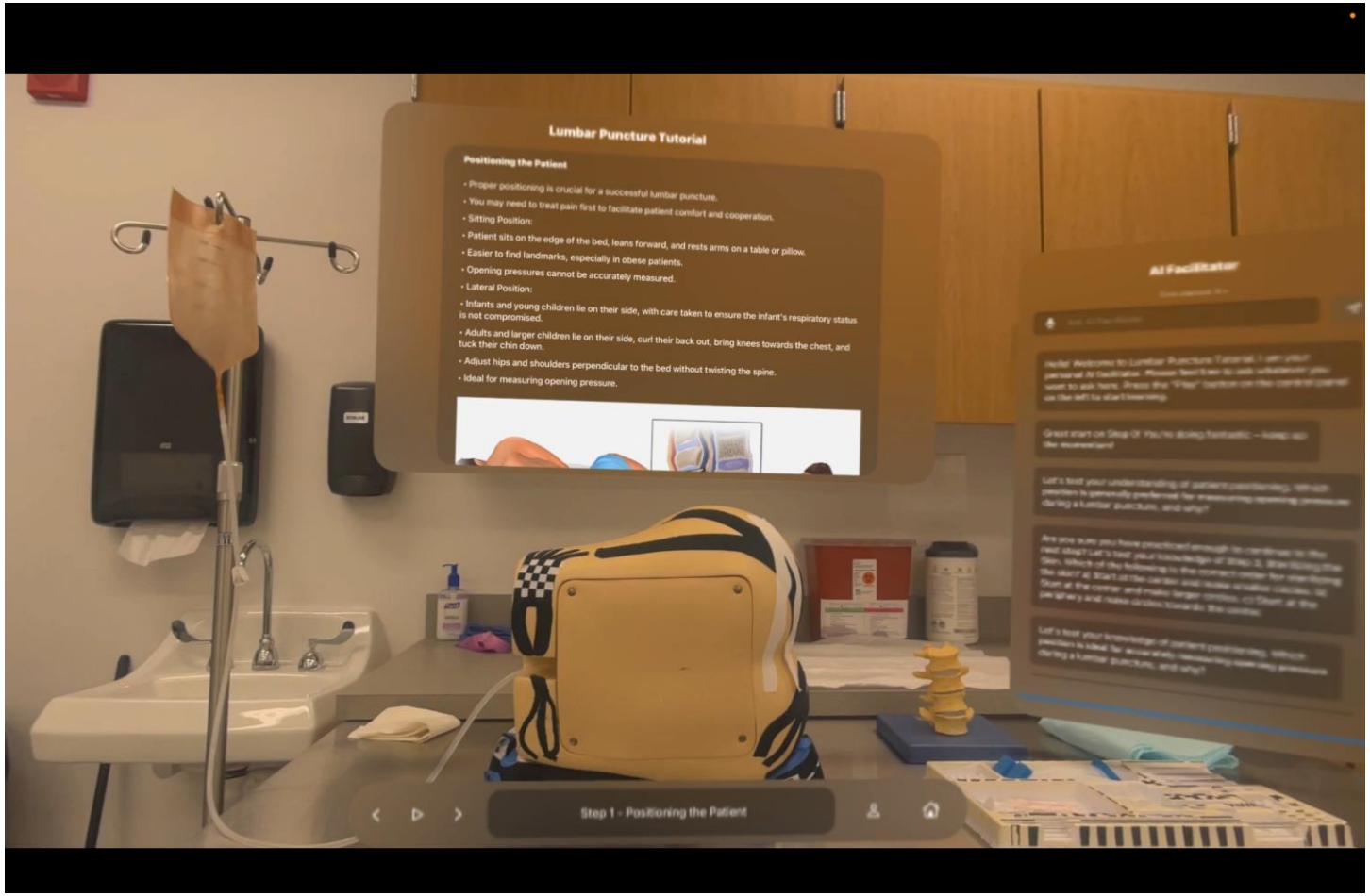
1. **Student often have only one chance to experience in-person simulation**
   - Need more practice to build skill.

2. **XR simulations allow independent, repeatable training**
   - Without needing a human facilitator present.

3. **AI can act as virtual facilitator**
   - Q/A, guidance, and support learning

Spatial AI Facilitator: Encouragement

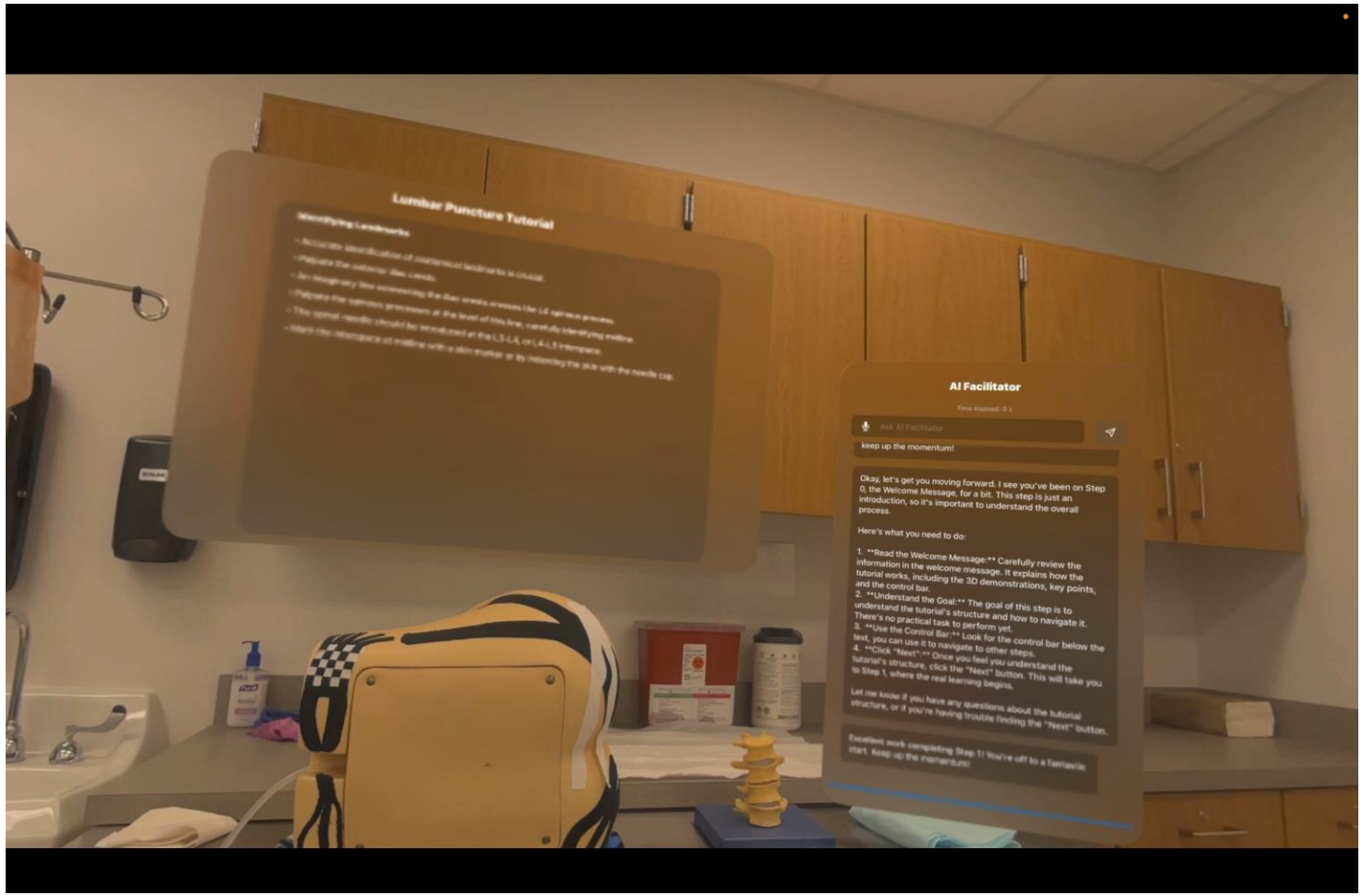
Carle Illinois COLLEGE OF MEDICINE



AI gives verbal encouragement and positive feedback

Spatial AI Facilitator: Too Fast

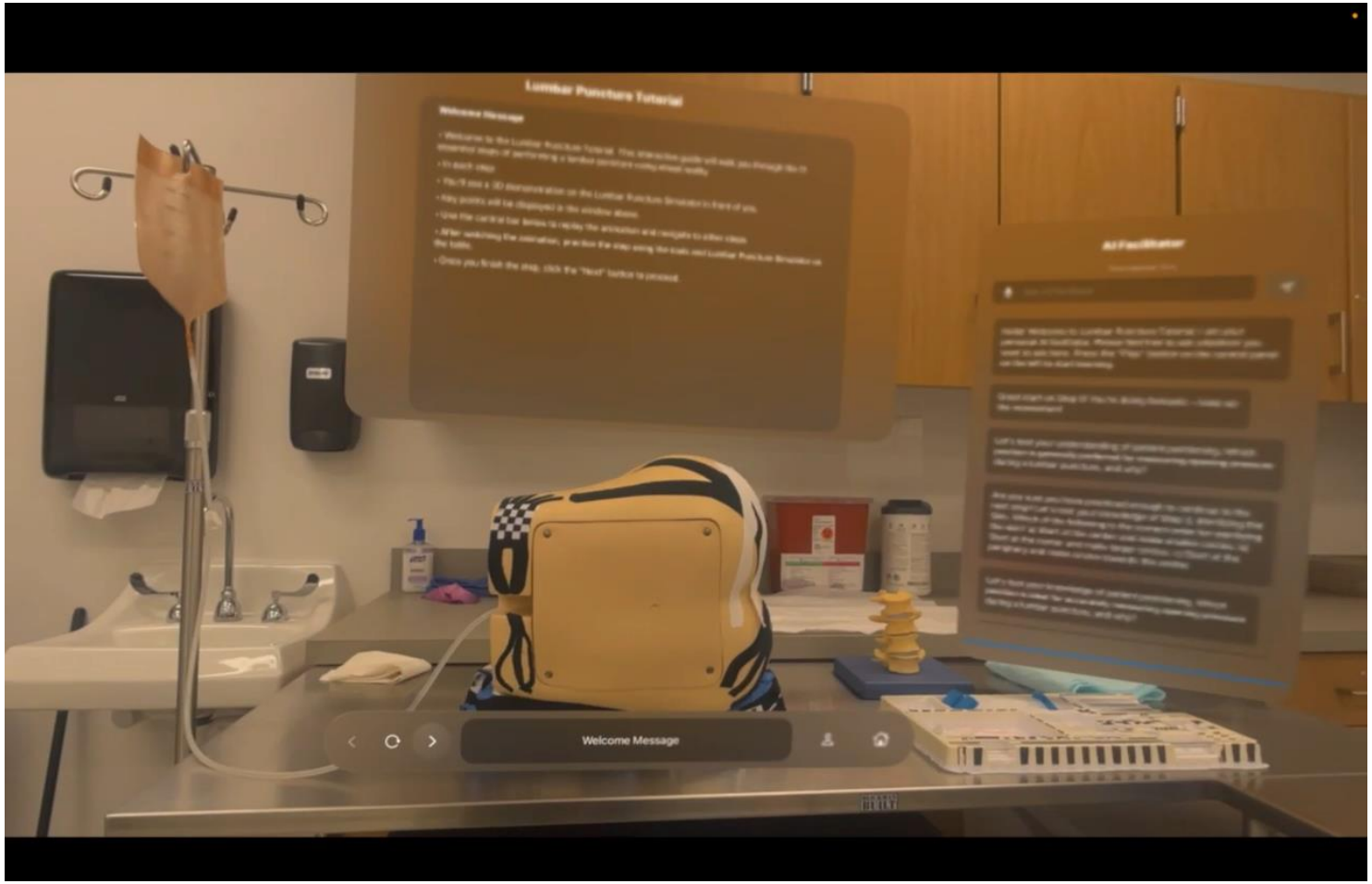
Carle Illinois COLLEGE OF MEDICINE



When learner moves too quickly through a critical step

Spatial AI Facilitator: Too Slow

Carle Illinois COLLEGE OF MEDICINE



When learner hesitates or gets stuck

Spatial AI Facilitator: User Study

Carle Illinois COLLEGE OF MEDICINE

Find the sweet spot where the AI facilitator is active enough to support learning – without interrupting or overwhelming the user

DEMO

jump
PowerDirector

Thank you!

Try out our demo at the VR Room!

EXAMPLE PROMPTS

Carle Illinois COLLEGE OF MEDICINE

What are your symptoms and how long have they been going on?

How did your pregnancy and delivery go?

How is your dog feeling today?

What is your family medical history?

How do you feel after intubation?

I'm going to put a nasal cannula on you, are you okay with that?